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[43]: %matplotlib inline
import matplotlib.pyplot as plt
plt.style.use('seaborn-whitegrid')
import numpy as np
import math
import random
import scipy
import scipy.linalg
with open('mat1-2.txt', 'r') as f:
    m = [[float(num) for num in line.split(',')] for line in f]
def randomList(numElements):
    myList = []
    for i in range(0,numElements):
        myList.append(random.uniform(-1,1))
    return myList
b=[]
for i in range(0,101):
    b.append(randomList(101))
errorsNormal=[]
errorsQR=[]
conditionNumbers=[]
for k in range(40,66):
    akList=[]
    averageNormalError=0
    averageQRError=0
    for i in range(0,len(m)):
        akList.append([])
        for j in range(0,k):
            akList[i].append(m[i][j])
    ak=np.array(akList)
    print("Size of A_40: %d x %d"%(k,len(ak),len(ak[0])))
    print("Rank of A_40: %d"%(k,np.linalg.matrix_rank(ak)))
    print("Condition Number of A_40: %d"%(k,np.linalg.cond(ak)))
    print()
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    for i in range(0,100):
        trueSol=np.array(np.linalg.lstsq(ak,b[i]))[0]
        #Cholesky raised an error for me so I used LU
        lower=scipy.linalg.lu(np.matmul(ak.T,ak))[1]
        upper=scipy.linalg.lu(np.matmul(ak.T,ak))[2]
        n=len(np.matmul(ak.T,ak))
        thisB=np.matmul(ak.T,b[i])
        y=np.linalg.solve(lower,thisB)
        normalSol=np.linalg.solve(upper,y)
        q,r=np.linalg.qr(ak)
        projection=np.dot(q.T,b[i])
        qrSol=np.linalg.solve(r,projection)
        normalError=np.linalg.norm(normalSol-trueSol)/np.linalg.norm(trueSol)
        qrError=np.linalg.norm(qrSol-trueSol)/np.linalg.norm(trueSol)
        averageNormalError+=normalError
        averageQRError+=qrError
    averageNormalError/=100
    averageQRError/=100
    errorsNormal.append(averageNormalError)
    errorsQR.append(averageQRError)
    conditionNumbers.append(np.linalg.cond(ak))
ks=[]
for i in range(40,66):
    ks.append(i)
```

Size of A_40: 101 x 40

Rank of A_40: 40

Condition Number of A_40: 74

Size of A_41: 101 x 41
Rank of A_41: 41
Condition Number of A_41: 103

Size of A_42: 101 x 42
Rank of A_42: 42
Condition Number of A_42: 152

Size of A_43: 101 x 43
Rank of A_43: 43
Condition Number of A_43: 217

Size of A_44: 101 x 44
Rank of A_44: 44
Condition Number of A_44: 328

Size of A_45: 101 x 45
Rank of A_45: 45
Condition Number of A_45: 483

Size of A_46: 101 x 46
Rank of A_46: 46
Condition Number of A_46: 753

Size of A_47: 101 x 47
Rank of A_47: 47
Condition Number of A_47: 1140

Size of A_48: 101 x 48

Size of A₅₀: 101 x 50
Rank of A₅₀: 50
Condition Number of A₅₀: 4695

Size of A₅₁: 101 x 51
Rank of A₅₁: 51
Condition Number of A₅₁: 7530

Size of A₅₂: 101 x 52
Rank of A₅₂: 52
Condition Number of A₅₂: 12789

Size of A₅₃: 101 x 53
Rank of A₅₃: 53
Condition Number of A₅₃: 21122

Size of A₅₄: 101 x 54
Rank of A₅₄: 54
Condition Number of A₅₄: 36949

Size of A₅₅: 101 x 55
Rank of A₅₅: 55
Condition Number of A₅₅: 62868

Size of A₅₆: 101 x 56
Rank of A₅₆: 56
Condition Number of A₅₆: 113329

Size of A₅₉: 101 x 59
Rank of A₅₉: 59
Condition Number of A₅₉: 668493

Size of A₆₀: 101 x 60
Rank of A₆₀: 60
Condition Number of A₆₀: 1282274

Size of A₆₁: 101 x 61
Rank of A₆₁: 61
Condition Number of A₆₁: 2395303

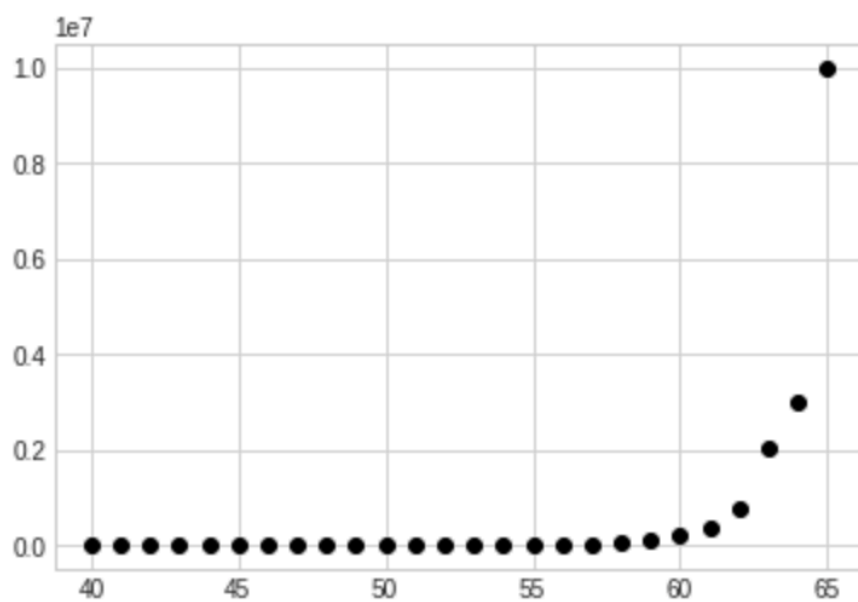
Size of A₆₂: 101 x 62
Rank of A₆₂: 62
Condition Number of A₆₂: 4745459

Size of A₆₃: 101 x 63
Rank of A₆₃: 63
Condition Number of A₆₃: 9161533

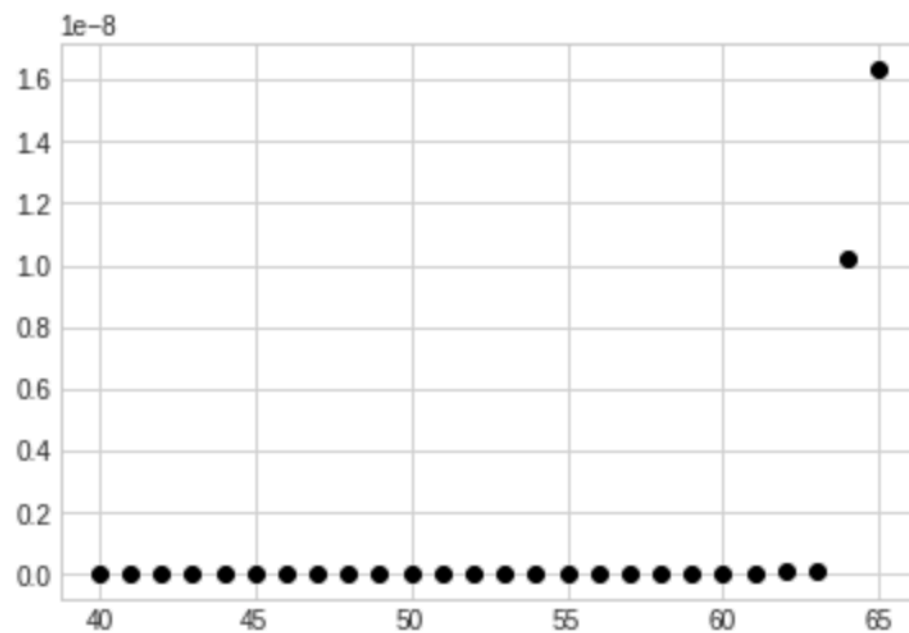
Size of A₆₄: 101 x 64
Rank of A₆₄: 64
Condition Number of A₆₄: 18765737

Size of A₆₅: 101 x 65
Rank of A₆₅: 65
Condition Number of A₆₅: 37486287

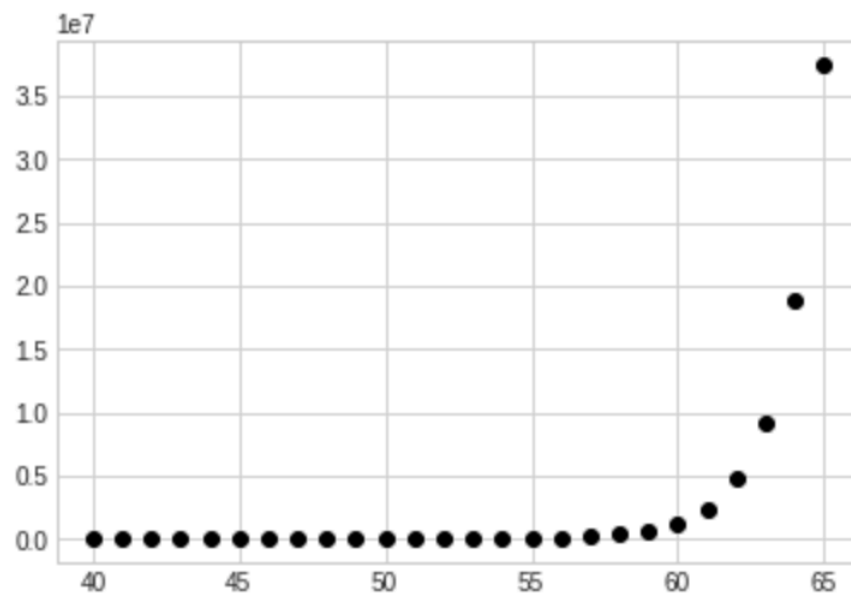
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[49]: plt.plot(ks, errorsNormal, 'o', color='black');
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[45]: plt.plot(ks, errorsQR, 'o', color='black');
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[50]: plt.plot(ks, conditionNumbers, 'o', color='black');
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1. The rate of increase in error as k increases appears to be much larger for the normal equations than the QR factorization.
2. The condition number appears to increase exponentially as k increases.

3. The QR decomposition is a much more favorable method for ill conditioned matrices because its rate of increase as k increases is smaller and the matrices become much more ill conditioned as k increases.